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Plasma proteomic biomarkers and the association between poor cardiovascular health and incident dementia: The UK Biobank study

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Abstract

Background: The study examined how plasma proteome indicators may explain the link between poor cardiovascular health (CVH) and dementia risk.

Methods: The present study involved 28,974 UK Biobank participants aged 50–74y (2006–2010) at baseline (2006–2010) who were followed-up for ≤ 15 y for incidence of dementia. CVH was calculated using Life's Essential 8 (LE8) total scores. The scores were standardized and reverse coded to reflect poor CVH (LE8z_rev). OLINK proteomics was available on this sample ($k=1,463$ plasma proteins). The study primarily tested the mediating effects of the plasma proteome in LE8z_rev-dementia effect. The total effect was decomposed into “mediation only” or pure indirect effect (PIE), “interaction only” or interaction referent (INTREF), “neither mediation nor

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†Performed statistical analyses.

CONSENT STATEMENT. The studies involving human participants were reviewed and approved by the UK Biobank which has approval from the Institutional Review Boards, namely, the North West Multi-center Research Ethics Committee for the United Kingdom, from the National Information Governance Board for Health and Social Care for England and Wales, and from the Community Health Index Advisory Group for Scotland. All participants gave informed consent for the study via a touch-screen interface that required agreement for all individual statements on the consent form as well as the participant's signature on an electronic pad. Written informed consent for participation was not required for this study in accordance with the National Legislation and the Institutional Requirements.

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interaction” or controlled direct effect (CDE), and “both mediation and interaction” or mediated interaction (INTMED).

Results: The study found poorer CVH assessed by LE8z_rev increased the risk of all-cause dementia by 11% [per 1 SD, hazard ratio, (HR)=1.11, 95% CI: 1.03–1.20, $p=0.005$]. The study identified 11 plasma proteins with strong mediating effects, with GDF15 having the strongest association with dementia risk (per 1 SD, HR=1.24, 95% CI: 1.16, 1.33, $P<0.001$ when LE8z_rev is set at its mean value) and the largest proportion mediated combining PIE and INTMED (62.6%; 48% of TE is PIE), followed by adrenomedullin or ADM. A first principal component with 10 top mediators (TNFRSF1A, GDF15, FSTL3, COL6A3, PLAUR, ADM, GFRAL, ACVRL1, TNFRSF6B, TGFA) mediated 53.6% of the LE8z_rev-dementia effect. Using all the significant PIE ($k=526$) proteins, we used OLINK Insight pathway analysis to identify key pathways, which revealed the involvement of the immune system, signal transduction, metabolism, disease, protein metabolism, hemostasis, membrane trafficking, extracellular matrix organization, developmental biology, and gene expression among others. STRING analysis revealed that five top consistent proteomic mediators were represented in two larger clusters reflecting numerous interconnected biological gene ontology pathways, most notably cytokine-mediated signaling pathway for GDF15 cluster (GO:0019221) and regulation of peptidyl-tyrosine phosphorylation for the ADM cluster (GO:0050730).

Conclusion: Dementia is linked to poor CVH mediated by GDF15 and ADM among several key proteomic markers which collectively explained ~54% of the total effect.

Keywords

Cardiovascular health; Life’s essential 8; plasma proteomic biomarkers; Dementia; Aging

INTRODUCTION

Healthy cognitive functioning to execute daily activities is ascribed to adequate attention, working or short-term memory, long-term memory, reasoning, movement coordination, and task-planning. The prevalence of brain illnesses that impact cognition, such as stroke and dementia, rises with age. Dementia is defined as the loss of global abilities in various cognitive domains, as well as the incapacity to execute everyday life activities. The estimated prevalence of dementia among persons over the age of 60 is 4.7% (Sosa-Ortiz et al., 2012) with 4.6–7.7 million as cases added each year globally [3.5–10.5 per 1,000] (Ferri et al., 2005; Helmer et al., 2006; Prince et al., 2013; Sosa-Ortiz et al., 2012). Patients, their proxies, and healthcare institutions are all greatly impacted by dementia (Ferri et al., 2005; Prince et al., 2013; Sosa-Ortiz et al., 2012). The most common form of dementia, Alzheimer’s disease (AD), accounting for 60–80% of cases (Sosa-Ortiz et al., 2012). AD is the primary cause of old age disability (Helmer et al., 2006) and has a higher health-care burden in developed countries (Helmer et al., 2006).

Dementia prevention is essential since there is currently no viable cure. The 2020 *Lancet commission* found that 40% of dementia risk can be attributed to early-, mid-, and later-life modifiable risk factors, such as education, hearing loss, traumatic brain injury, hypertension, alcohol use, obesity, smoking, depression, social isolation, physical inactivity, air pollution,

and diabetes, despite the partial genetic basis of late-onset AD (e.g., ApoE ϵ 4) (Livingston et al., 2020). For prevention and designing cost-effective therapies, identifying new mid-life risk factors and pathways is essential. Life's Essential 8, a composite score that combines many cardiovascular health (CVH) parameters (Hayman and Martyn-Nemeth, 2022; Lloyd-Jones et al., 2022) was recently proposed as an important effect for a number of health outcomes, including the incidence of dementia (Beydoun et al., 2023a; Petermann-Rocha et al., 2023; Wang et al., 2023; Zhou et al., 2023).

Using data from the UK Biobank cohort, several recent studies have indicated that LE8 total score, with higher score indicative of better CVH, was inversely related to dementia incidence, perhaps differentially by AD genetic risk as measured by the ApoE ϵ 4 status (Zhou et al., 2023), and that its lifestyle sub-score composed of diet, physical activity, smoking and sleep partially mediated some of the socio-economic disparities in dementia risk (Beydoun et al., 2023a). Other studies using various other cohorts, also indicated that LE8 total score, as well as its predecessor Life's Simple 7, LS7 (Hayman and Martyn-Nemeth, 2022; Lloyd-Jones et al., 2010), which among others excluded sleep health from its final score, may be protective against various adverse cognitive outcomes, including age-related cognitive decline and incidence of dementia (Atkins et al., 2019; Gildner et al., 2018; Gonzalez et al., 2016; Gonzalez et al., 2018; Gorelick et al., 2017; Guo et al., 2021; Hessler et al., 2016; Jeon et al., 2021; Malik et al., 2021; Ogunmoroti et al., 2016; Samieri et al., 2018; Thacker et al., 2014; Tin et al., 2022; Wei et al., 2022). Studies on the plasma proteome and CVH and its components have largely focused on a narrow subset of proteins. The plasma proteome has been related to all-cause dementia, on the other hand, using targeted and large-scale proteomics (Harris et al., 2020; Lindbohm et al., 2022; Tanaka et al., 2020; Trares et al., 2022; Whelan et al., 2019).

In this retrospective cohort study, we assessed poor CVH in relation to the incidence of all-cause dementia using data from the UK Biobank, while attempting to explain apparent relationships through the plasma proteome, because most studies have addressed a limited number of proteins or small cohort sizes.

MATERIALS AND METHODS

Database

The UK Biobank is a prospective study of about 500,000 persons aged 37–73 years at the time of recruitment who lived in the UK between 2006 and 2010 (UK Biobank, 2007). The purpose and design of the study are described elsewhere (UK Biobank, 2007). Participants were recruited and sent to one of 22 assessment centers (within 25 miles) in England, Scotland, or Wales to complete self-administered and touch-screen questionnaires, as well as a face-to-face interview (UK Biobank, 2007). Biological samples and phenotypic measures were obtained (UK Biobank, 2007). The UK Biobank questionnaire selected a broad variety of measurable exposures in a wide range of interest areas after a rigorous evaluation of previous observational studies, clinical trials, and demographic surveys, as well as consultation with international experts (UK Biobank, 2007). The Northwest Multi-Centre Research Ethics Committee granted ethical approval to the UK Biobank initiative. The

present application (#77963) was also approved by the UK Biobank access management team and by the National Institutes of Health's Institutional Review Board.

Dementia outcomes: Using algorithmically generated data for dementia occurrences that relied mainly on hospital registry linkage (reported in fields 42018 and 42020; described in: https://biobank.ndph.ox.ac.uk/showcase/showcase/docs/alg_outcome_main.pdf), we removed participants whose age at the start of dementia was younger than their initial age at baseline assessment (UK Biobank, 2022). We used the ICD-10 codes F00 or G30 to identify new instances of Alzheimer's disease (AD). However, for dementia of any etiology, including vascular dementia (F01, I67.3), a variety of codes were used, including A81.0, F00, F01, F02, F03, F05, G30, G31.0, G31.1, G31.8, and I67.3. Using the previously described algorithm (UK Biobank, 2022), the date of the first incident of any form of dementia was determined by picking the earliest date from several accessible date variables/fields associated to this result.

Cardiovascular health: Life's essential 8—In 2010, the American Heart Association (AHA) broadened its scope of interest by developing a new measure of cardiovascular health (CVH) aimed at individual and population-level health promotion (Hayman and Martyn-Nemeth, 2022; Lloyd-Jones et al., 2010). CVH was described as seven potentially modifiable biological and behavioral parameters that, when at optimal levels, result in increased CVD-free survival, longevity, and quality of life. This CVH measure was dubbed “Life's Simple 7” (LS7), and its seven components included indicators of diet quality, increased physical activity, decreased cigarette smoking, lower BMI, total cholesterol, fasting blood glucose levels, and optimal blood pressure levels. Each statistic was categorized as poor (0), middle (1), or perfect (2) using clinical thresholds that were accepted for both children and adults. LS7's total summary score can range from 0 (all metrics at terrible levels) to 14 (all seven measures at optimal levels) (Hayman and Martyn-Nemeth, 2022; Lloyd-Jones et al., 2010). Since the publication of the 2010 AHA statement, CVH has been re-evaluated, and an AHA Presidential Advisory presented an improved version of CVH, reflecting gains achieved over a decade of research, while LS7's methodological limitations have been addressed. This new measure was dubbed “Life's Essential 8” (LE8), and it retained all seven components of LS7 while making significant changes to definitions and scales. Given its documented influence on CVH over the life span (Hayman and Martyn-Nemeth, 2022; Lloyd-Jones et al., 2022), sleep health was incorporated into the life's essential 8 (LE8) total score. This measure is described in detail in online supplementary method 1 [OSM1, supplementary Tables 1–3 (Appendix II)]. The total LE8 score normally can range from 0 to 800 with a higher score implying better CVH. In this study, the score was z-standardized and reverse coded (by multiplying the z-scored by -1) or $LE8_{z_rev}$ such that a higher score would indicate poorer CVH and 1 unit increase indicates a 1 standard deviation (SD) decrement in LE8 score. The same was done individually to the LE8 lifestyle sub-score, which included the components of diet, physical activity, smoking, and sleep and the LE8 biological sub-score, which included the components of body mass index, blood lipids, blood glucose, and blood pressure. LE8 was determined at the baseline assessment for UK Biobank participants with data on its components.

OLINK proteomics: A large-scale plasma proteome study was one of the components of the UK Biobank Pharma Proteomics Project (UKB-PPP). The Olink® Explore 1536 Proteomics platform was used to analyze 54,306 plasma samples from individual UK Biobank participant-visits. This platform employs Proximity Extension Assay (PEA) technology to quantify 1,472 protein analytes, representing 1,463 distinct proteins from inflammation, cancer, cardiometabolic, and neurological panels. Sun et al. (Sun et al., 2023) provide a complete explanation of the UKB-PPP implementation of PEA assays in this cohort, as well as data pre-processing and normalization. The protein data is presented in log2 normalized protein expression (NPX) units (OSM 2, Appendix II). This study uses proteomic data from the baseline assessment of the UK Biobank study.

Covariates: Potential socio-demographic confounders included age, gender, race/ethnicity (White, Black, South Asian, and Others), and household size. Educational attainment, household income, and the Townsend deprivation index (TDI) were identified as socioeconomic status indicators. A touch-screen questionnaire collected baseline data on educational attainment, which was re-grouped based on previous research²⁰ as 0 = Low, combining “None”, “CSEs/Equivalent”, “NVQ/HND/HNC/Equivalent” and “Other professional qual”; 1 = Intermediate, combining “O Levels/GCEs/Equivalent” and “A/AS Levels Equivalent”; 2 = Higher level or “College/University.” Total household income before tax was divided into five categories: 1=“£18,000,” 2=“£18,000-£29,999,” 3=“£30,000-£51,999,” 4=“£52,000-£100,000,” and 5=“>£100,000.” TDI ratings were computed using national census statistics, focusing on residential postcode-level car ownership, home overcrowding, owner occupation, and unemployment. Higher TDI scores indicate more socioeconomic deprivation (Townsend P, 1987). As a result, the TDI was multiplied by -1 to signify higher SES and combined with z-scores of educational attainment and household income to provide a single SES summary score. Finally, Alzheimer’s Disease Polygenic Risk Score (AD PRS) was considered as a potential effect modifier in part (See OSM 3, Appendix II).

Study sample selection—The UK Biobank sample consisted of 502,268 participants, with 384,548 aged ≥50 years at baseline. 270,579 had available data on socio-demographic, SES, lifestyle, and biological factors. Only 10% had proteomic data, resulting in 29,010 participants with complete plasma protein data. The final sample size was 28,974 participants after excluding 36 prevalent dementia cases, with 786 having incident all-cause dementia over 15 years (Appendix III, Figure S1). A sub-set of 28,801 participants had complete AD PRS data and were included in secondary analyses stratified by AD PRS tertiles.

Statistical methods: Stata 18.0 (StataCorp, College Station, TX) was used for all analyses. We started with some descriptive analysis. Means and proportions of major study variables were determined in the whole chosen sample, with sex stratification. Using linear, logistic, and multinomial logit models, these estimates were also compared by gender.

Second, without adding the mediator at this step, exposure-outcome connections were examined in the final chosen analytic sample with full mediator data. To that purpose, time-to-event was defined as the interval between age at entrance into (≥ 50y) and age at exit

from the study, which was defined as either the age at incident or the age at censoring (death or end of follow-up set to October 31st, 2021). To demonstrate variations in dementia-free survival probability, Kaplan-Meier survival rates during age of follow-up were computed and compared using a log-rank test across the tertile version of exposure of interest, LE8z_rev: T₁ (better CVH); T₂ (intermediate CVH); or T₃ (poor CVH). Following that, we built Cox proportional hazards (PH) models to examine the correlations of LE8z_rev tertiles with all-cause dementia while controlling for baseline age, sex, racial minority status (NonWhite vs. White), household size, and SES z-score. Assumptions for the main Cox PH model were verified using Schoenfeld residuals and other methods that determine deviation from the proportionality of hazards assumption.

Third, the LE8z_rev exposure was entered as the main predictor into a series of separate multiple linear regression models, alternating each proteomic mediator as the outcome of interest in this analysis, among 1,463 plasma proteomic biomarkers, with simultaneous adjustment for the same covariates mentioned above (Stata *parmby*, *qqval*, and *multproc* commands). A volcano plot was generated using the R *ggplot* package to illustrate the p-values and effect sizes of each of the 1,463 equations and to highlight those that passed multiple testing with the Bonferroni correction. The top hits which were selected had >0.20 effect size in absolute value (corresponding to 1/5 SD higher plasma protein per SD decrement in LE8 total score), a value known to be a cutoff between very weak (or very small) and weak (or small) effect size when considering the association between two continuous variables, using several standards including those applied to Cohen's D (<https://cran.r-project.org/web/packages/effectsize/vignettes/interpret.html>). This step was only included if the Bonferroni correction yielded a large number of potential mediators (e.g. >500). A similar approach was adopted in a previous study (Beydoun et al., 2023b).

Fourth, the selected proteome biomarkers were included into a 4-way decomposition model with LE8z_rev as the exposure variable, time to dementia as the outcome variable, and one of each chosen proteomic biomarker as a potentially mediating variable, using a Cox PH model as the final equation. Furthermore, these proteomic biomarkers were allowed to act as possible moderators in the link between exposure and outcome factors. The four-way decomposition is explored in further depth in OSM4 (Appendix II), and a technical appendix (Appendix IV) is also given. The results are presented in a tabular format with heatmap visualizations only for the selected key mediators that passed Bonferroni correction and other selection criteria above, separating “no mediation” from “consistent” (controlled direct effect (CDE)< total effect (TE) in absolute value, sign of the pure indirect effect (PIE) is the same as for CDE) and “inconsistent mediation” (CDE>TE in absolute value, PIE and CDE have opposing signs) (Discacciati et al., 2018; Karriker-Jaffe et al., 2011; Pamplin Ii et al., 2023). According to their function and link to CVH risk factors, LS7, LE8 and dementia in previous research, the top identified mediators or moderators of the LE8z_rev -dementia association are emphasized.

Fifth, if any of the pre-selected plasma proteins were statistically significant mediators with a significant PIE at 0.05 type I error, they were included in a principal components analysis (PCA) for dimension reduction, as long as TE was statistically significant at a type I error of 0.05 and there was a consistent mediation (CDE<TE in absolute value, and PIE has the same

sign as CDE). When many components were retrieved, they were orthogonally rotated using varimax for ease of interpretation. The number of extracted components was determined using the Kaiser rule (eigenvalue >1). These PCA scores (z-scores) were predicted using the regression technique and integrated into another set of 4-way decomposition models to estimate the degree of mediation and/or moderation by these components in the overall impact of LE8z_rev on incident dementia. In most of our analyses, sex served as the dominant stratifying factor and regression models were conducted on the total sample as well as separately on men and women. In this particular analysis examining the PCA score in a four-way decomposition model, AD PRS tertiles were also considered as a secondary stratifying variable. As a secondary analysis, top mediators (consistent and/or inconsistent) were also visualized in relation to the main exposure, its lifestyle and biological sub-scores and its individual components using several heatmaps as appropriate and similar OLS models that adjusted for age, sex, racial minority status, SES and household size.

Sixth, as another secondary analysis, the supplementary materials provide results of four-way decomposition throughout the proteome (k=1,463 proteins), by extracting and listing the plasma proteins with statistically significant TE and PIE at a type error of 0.05. No Bonferroni correction was made for this secondary analysis and a similar approach was used in an earlier study (Beydoun et al., 2023b). These findings were sent into the OLINK insight pathways browser (<https://insight.olink.com/>) to determine the most often utilized paths by those mediators (<https://github.com/baydounm/UKB-paper11-supplementarydata>), which was then shown as a collection of independent and related pathways. Furthermore, the important mediators with statistically significant PIE were subjected to Search Tool for the Retrieval of Interacting Genes/Proteins (STRING) analysis (<https://string-db.org>) and k-means clustering with up to 10 clusters. This offered further information on other proteomic biomarkers identified as important mediators based on PIE, as well as their clustering or functional connections with the top mediators. Gene ontology (GO) pathways within these clusters were described and top pathways were determined using a combination of the lowest false discovery rate with a strength value >1. Strength in GO reflects the level of confidence or support for the functional relationship between different genes or terms based on available biological evidence.

Data Availability Statement: The following licenses/restrictions apply to the data analyzed in this study: UK Biobank is a large-scale biological database and research resource that contains detailed genetic and health information from over 500,000 UK individuals. The database is frequently updated with new information and is available to approved researchers worldwide who are conducting critical research into the most common and life-threatening diseases. Access to these datasets should be requested through <https://www.ukbiobank.ac.uk/>.

RESULTS:

In this study sample of 28,974 UK Biobank individuals aged 50 or older, we found that factors such as age, race, household size, SES index, and LE8 total and sub-scores varied by sex (Table 1). Women had higher LE8 total, lifestyle and biological sub-scores, suggesting better CVH, but lower SES. Poor CVH significantly increases the risk of dementia in males

over time, based on the Kaplan-Meier curves, with the relationship mainly observed in the contrast between T₃ and T₁ of LE8z_rev (Figure 1). Mean follow-up time was 12.1 y from 2006–2010 (baseline assessment years) up to October 31st, 2021, with an incidence rate of 2.2 per 1,000 person-years. During this follow-up time, 3,088 deaths occurred of the total selected sample (N=28,974). Comparing the final sample that was selected to the unselected sample that was aged 50+y at baseline assessment, being selected was more likely at older ages, among men, and at lower SES groups, in a multivariable logistic regression model with selection status as the outcome which included age, sex, race (Non-White vs. White), SES z-score and household size as the main predictors. When sample selectivity was examined using bivariate analyses, most notably, the selected sample had 52% females vs. 54% in the unselected sample ($P<0.001$, chi-square test), (Output provided in <https://github.com/beydounm/UKB-paper11-supplementarydata>).

Table 2 shows the overall association of LE8z_rev with incident dementia, before and after stratifying by sex and AD PRS tertile, using a series of Cox proportional hazards models that were adjusted for age, sex, race/ethnicity, household size, and SES. Among both sexes, LE8z_rev (each SD decrement in LE8 total score, equivalent to 93 points of 800), was associated with incident dementia (HR=1.11, 95% CI: 1.03, 1.20), and especially in the lowest tertile of AD PRS (HR=1.42, 95% CI: 1.17–1.72), with association weakening in a linear manner with increased AD genetic risk ($P=0.006$ for 2-way interaction between AD PRS tertile and LE8z_rev). After stratifying by sex, the relationship between LE8z_rev and incident dementia was detected only among men (HR=1.12, 95% CI: 1.01, 1.24) with a similar heterogeneity by AD genetic risk ($P=0.001$ for the 2-way interaction between AD PRS tertile and LE8z_rev).

A series of 1,463 multiple ordinary least square regression models were conducted to test the strongest associations between LE8z_rev, the key exposure of interest, and the plasma proteome (i.e. the set of potential mediators between LE8z_rev and dementia). These models were adjusted for the same set of covariates as above. A total of 1,252 plasma proteins were significantly predicted by LE8z_rev after Bonferroni correction. Among those, only 144 had an effect size β that was either <-0.20 or $>+0.20$, reflecting a change of 1/5 standard deviation in plasma protein with each SD increase in LE8z_rev, as visualized in Figure 2. Selected strongly associated proteins ($k=144$) were included into a four-way decomposition models with the exposure being LE8z_rev and the outcome being incident dementia modeled with Cox regression. In Figure 2, the strongest positive effect sizes were found for the protein leptin (LEP), closely followed by FABP4 and LDLR; while on the other side, the strongest inverse effect size was found for IGFBP2.

Table 3, Figure 3 and Appendix V (supplementary datasheet 1) present key findings of four-way decomposition models, adjusted for the same set of covariates listed above, as applied to the selected proteins with strong association with the main exposure ($k=144$). The findings were classified into three group as shown in Figure 3: **(A)** no mediation ($P>0.05$ for PIE); **(B)** Inconsistent mediation ($P<0.05$ for PIE with $|CDE|>|TE|$ and PIE sign opposite to that of TE); **(C)** Consistent mediation ($P<0.05$ for PIE with $|CDE|<|TE|$ and PIE sign the same as that of TE). Detailed findings for consistent mediation are presented in Table 3 and complete results are presented in Appendix V (supplementary datasheet

1). Findings indicated that most of the selected proteins were not significant mediators ($k=77$: i.e. **group A**), while another large portion were inconsistent mediators ($k=56$, i.e. **group B**) and only 11 were consistent mediators (i.e. **group C**). Among those consistent mediators, the growth differentiation factor 15 (GDF15, Log2 transformed, z-scored) had the strongest association with dementia risk (per 1 SD, HR=1.24, 95% CI: 1.16, 1.33, $P<0.001$ when LE8z_rev is set at its mean value of ~498 points on the LE8 total score scale) and the largest proportion mediated combining PIE and INTMED (62.6%; 48% of TE is PIE), closely followed by adrenomedullin or ADM (56%; 41% of TE is PIE). Other plasma proteins deemed consistent mediators explained over 20% but <40% of TE as PIE, and those included GFRA1, ACVRL1, COL6A3, PLAUR, and TGFA. All other selected plasma proteins found to have consistent mediating effect on the CVH-dementia association had a proportion mediated <20% as PIE (TNFRSF1A, CA14, FSTL3 and TNFRSF6B). Among the most influential inconsistent mediators were FURIN and IGFBP2, with >30% suppressive effect (**group B**). A detailed literature review of the selected consistent mediator plasma proteins, including GDF15 and ADM among others, in terms of their association with dementia traits and CVH is presented in OSM6 and supplementary Table 5.

A principal components analysis (PCA) was conducted in which a limited number of components were extracted using the Kaiser rule (eigenvalue>1) and was applied to those top 11 significant proteomic mediators belonging to group C (i.e. consistent mediators), (details provided in OSM5 and supplementary Table 4, Appendix II). Consequently, two principal components were extracted (PC1 and PC2) (supplementary Figure 2, Appendix VI). PC1 reflecting mainly variance in 10 of these top mediators (excluding CA14), whereas PC2 mainly reflected variance in CA14. When entered into a four-way decomposition model for poor CVH vs. dementia, adjusting for the same set of covariates as above, PC1 explained 53.6% of the LE8z_rev-dementia effect (95% CI: 13.7%–93.4%), increasing dementia risk by 6% and a standardized effect size with LE8z_rev $b=0.75$. Similarly, PC2 explained around 23% of the total effect of LE8z_rev on dementia risk, as PIE, with an overall percent mediated estimated at 19.8%. PC1 mediated the LE8_{lifestyle} and LE8_{biological} z-scored and reverse-coded sub-scores' association with dementia risk similarly across sex and AD PRS groups. In contrast, mediation was mostly noted in the second AD PRS tertiles in the case of PC2 for all 3 alternative exposures. The findings in the overall sample and stratified by sex and AD PRS tertiles are presented in Appendix VII (supplementary Figure 3) and its associated datasheet and Output available on <https://github.com/baydounm/UKB-paper11-supplementarydata>.

Approximately 1,271 pathways were found overall from our OLINK insight pathway analysis, which included all statistically significant mediators ($k=526$ proteins, Appendix VIII (supplementary Figure 4). The domains of the immune system, signal transduction, metabolism, disease (including infectious and metabolic diseases), protein metabolism, hemostasis, membrane trafficking, extracellular matrix organization, developmental biology, and gene expression (transcription) were among covered top pathways. Appendix IX (supplementary datasheet 2) has detailed pathways ($p=1,271$). The following link contains all of the supplemental datasheets, comprehensive code, and relevant outcome datasets that were utilized to create the tables and figures: <https://github.com/baydounm/UKB-paper11->

[supplementarydata](#). In addition, GDF15 was identified in cluster #1 (of the 526 plasma proteins with a substantial PIE) according on the STRING analysis, along with 48 other mediators. Only one of the top mediators, GDF15, was chosen for the PCA. Conversely, cluster #8 (63 proteins), which contained ADM as one of the top mediators, also contained three additional top mediators—TNFRSF1A, PLAUR, and GFRA1—that were included in the PCA. Appendix VIII (Supplemental Figure 4) displays both clusters as well as the cluster analysis result and the outcomes for all mediators. Thus, when all mediators were taken into account, two of the largest plasma protein clusters also contained five top mediators which were included in the PCA (See KEY_MEDIATORS.txt supplementary file for full list of mediators; Appendix X). Gene ontology (GO) and KEGG analyses are also provided for clusters #1 and #8 and stored in the github repository: <https://github.com/baydounm/UKB-paper11-supplementarydata>. More specifically, these two larger clusters reflected numerous interconnected biological GO pathways with the lowest false discovery rate (FDR) coupled with highest strength (>1) ascribed to cytokine-mediated signaling pathway (GO:0019221) and regulation of peptidyl-tyrosine phosphorylation (GO:0050730) for each of these two clusters containing GDF15 and ADM, respectively. Appendix XI (Supplementary Figure 5) visualizes the main findings.

In a secondary analysis, top 11 consistent mediators and top 56 inconsistent mediators were included as alternative outcomes in a series of OLS linear regression with main predictors being alternatively the main exposure (LE8z_rev_total), LE8z_rev_lifestyle, LE8z_rev_biological and Log_e transformed, z-scored and reversed (multiplied by -1) components of the LE8 (i.e. 8 components), adjusting for age, sex, racial minority status, SES and household size. The effect sizes are visualized as heatmaps in Appendix XII (supplementary Figure 6) and the findings are stored under [supplementary datasheet 3 \(Appendix XIII\)](#) in <https://github.com/baydounm/UKB-paper11-supplementarydata>.

DISCUSSION:

This retrospective cohort study using UK Biobank data found that poor CVH is linked to an 11% increase in the risk of all-cause dementia. The study identified 11 plasma proteins with strong mediating effects, with GDF15 having the strongest association with dementia risk and the largest proportion mediated combining PIE and INTMED, followed by adrenomedullin or ADM. A principal component with 10 top mediators mediated 53.6% of the LE8z_rev-dementia effect. OLINK Insight pathway analysis revealed the involvement of the immune system, signal transduction, metabolism, disease, protein metabolism, hemostasis, membrane trafficking, extracellular matrix organization, developmental biology, and gene expression among others. STRING analysis revealed five top consistent mediators in two larger clusters reflecting numerous interconnected biological GO pathways, most notably cytokine-mediated signaling pathway for GDF15 and regulation of peptidyl-tyrosine phosphorylation for ADM.

Numerous studies have examined the relationship between LE8, LS7 or individual components of LE8 with dementia incidence (Atkins et al., 2019; Gildner et al., 2018; Gonzalez et al., 2016; Gonzalez et al., 2018; Gorelick et al., 2017; Guo et al., 2021; Hessler et al., 2016; Jeon et al., 2021; Malik et al., 2021; Ogunmoroti et al., 2016; Samieri et

al., 2018; Thacker et al., 2014; Tin et al., 2022; Wei et al., 2022). For instance, a study involving 296 dementia cases found that smoking, poor physical activity, and poor fasting glucose levels were associated with an increased risk of dementia. However, body mass index, blood pressure, and cholesterol were not meaningful predictors of dementia (Hessler et al., 2016). Vitamin D was found to be an effect modifier in the association between CVH and cognition, especially in men in another study (Jeon et al., 2021). Adherence to Life's Simple 7 (LS7) metrics was associated with better cognitive performance based on a recent analysis of the NHANES study (Wei et al., 2022). Another study in France found that increased numbers of optimal CVH metrics and a higher CVH score were associated with a lower risk of dementia and cognitive decline in older persons (Samieri et al., 2018). Additionally, several meta-analyses of observational cohort studies add evidence of an association between poor diet, physical inactivity, and other lifestyle and biological components of LE8 in relation to dementia risk [e.g. (Liu et al., 2020; Zhang et al., 2023)].

Eleven proteins were deemed top consistent mediators between poor CVH and dementia risk, namely TNFRSF1A, CA14, GDF15, FSTL3, COL6A3, PLAUR, ADM, GFRA1, ACVRL1, TNFRSF6B, TGFA (see supplementary Table 5 for details). In particular, the pleiotropic factor GDF15 controls cellular stress responses and inflammation. In particular, the pleiotropic factor GDF15 was a top mediator. GDF15 plays a role in cellular stress responses and inflammation and recent data links GDF15 to inadequate CVH. (Beydoun et al., 2023c; Jiang et al., 2021; Reyes and Yap, 2023). ADM is another important mediator, accounting for more than 40% of the overall impact of poor CVH on dementia as determined by the LE8 total score (z-scored and reverse coded). The ADM gene encodes a preprohormone, cleaved into adrenomedullin and proadrenomedullin N-terminal 20 peptide, that functions in vasodilation, hormone regulation, angiogenesis, and specific antimicrobial activity (<https://www.ncbi.nlm.nih.gov/gene/133>). ADM interferes with microtubule dynamics and may intervene in neuronal dysfunction through immune response, inflammatory response, apoptosis, or calcium dyshomeostasis. Thus, it is now a drug target for many neurodegenerative diseases (Ferrero et al., 2018). Consequently, recent data supports our findings by indicating associations between poor CVH and dementia possibly through GDF15, ADM, and other plasma proteins. Nevertheless, replication by multiple cohorts and additional mechanistic studies are needed to confirm the causal role played by these selected proteins.

CA14 was a singular plasma protein with a strong component loading on a second principal component in our PCA analysis for top mediators. CA14 was found to be inversely related to poor CVH as well as to dementia risk. Carbonic anhydrases (CAs) are zinc metalloenzymes that reversibly hydrate carbon dioxide, and are involved in various biological activities like respiration, calcification, and bone resorption. CAs vary in tissue distribution and localization, serving different physiologic roles, including the formation of the cerebrospinal fluid (<https://www.ncbi.nlm.nih.gov/gene/23632#gene-expression>). A very recent study tested carbonic anhydrase inhibitors (CAIs) methazolamide and acetazolamide against Alzheimer's disease (AD) and cerebral amyloid angiopathy (CAA) in mice (Canepa et al., 2023). Results showed that CAIs reduced A β accumulation, gliosis, and improved cognition and microvascular fitness, while reducing A β deposition. The study supported the potential application of CAIs in clinical trials for AD and CAA (Canepa et al., 2023).

With a few notable exceptions in recent research, studies on the plasma proteome and CVH and its constituents have primarily concentrated on a small selection of proteins. Several common findings were detected in our present analysis when compared with those previous large-scale studies. For instance, Tanaka and colleagues found that GDF15 and IGFBP2 were among the strongest mediators between poor CVH as measured by LS7 and all-cause mortality, with leptin being the most strongly associated with poor CVH (Tanaka et al., 2023). These findings were very similar to our work even though our outcome was incident all-cause dementia (Tanaka et al., 2023). We also found that one of the top hits with poor CVH but not dementia was leptin. There is evidence that the adipokine hormone leptin, which was first associated with hunger and eating habits, acts pleiotropically in the cortical and hippocampus, areas of the brain involved in cognition (Farooqi et al., 2007; Jacob et al., 1997; Zhang et al., 1994). In fact, it has been demonstrated that giving mice leptin directly into the hippocampus enhances memory processing and shapes the hypothalamus from an early age (Harvey et al., 2005). Many recent epidemiological studies have found an association between leptin and various brain health outcomes (Gunstad et al., 2008; Holden et al., 2009; Paz-Filho et al., 2008). Although established neuroanatomical connections suggest an interaction between areas influencing energy homeostasis and cognition, this is still not well understood (Morrison, 2009). Our findings do not indicate that leptin is a key mediator in the association between poor CVH, which includes adiposity, and dementia risk.

Our finding of a shared cytokine-mediated signaling pathway between dementia and CVH, specifically for the GDF15 cluster, is a replication of at least one other study that detected the same GO pathway (GO:0019221) in both AD and type 2 diabetes among others (Castillo-Velazquez et al., 2023). The other strong pathway that we detected was regulation of peptidyl-tyrosine phosphorylation for the ADM cluster (GO:0050730). Tyrosine kinase inhibitors were among the least investigated pharmaceuticals among those repurposed treatments for dementia (Heard et al., 2018). Curcumin, containing significant amounts of tyrosine kinase inhibitors (Golonko et al., 2019), has recently been revealed in a meta-analysis of large controlled clinical studies to reduce age-related cognitive decline, including in the domain of working memory (Tsai et al., 2021). Curcumin has also been linked to improvement in dementia-related brain pathology (Reddy et al., 2018). Nonetheless, more trials on a varied collection of medicines and nutraceuticals known to be tyrosine kinase inhibitors for dementia prevention are required, pending replication of these pathway findings in other cohort studies.

More importantly, numerous selected plasma proteins ($k=56$ of the 144 top proteins) were found to be inconsistent mediators, whereby manipulating their values to a lower level would increase the effect of poor CVH on dementia risk. Some of the top inconsistent mediators with the largest percent mediated and with the strongest effects on LE8 total score exposure included furin (on the positive side) and IGFBP2 (on the negative side), (Figure 2). Previous animal and human studies show that elevated furin (Choi et al., 2021; Zhang et al., 2022) and reduced IGFBP2 (McGrath et al., 2019) were inversely associated with adverse brain health outcomes, thus replicating our present findings in a large observational study of humans. Examples of the inconsistent mediation in the literature are described in the technical appendix on four-way decomposition for ease of interpretation (Appendix IV). Thus, experimental manipulation of these proteins would be counterproductive in preventing

dementia occurrence ascribed to poor CVH and primary focus should be on consistent mediators as described earlier.

Our current research has both notable strengths and weaknesses. First, this research examined a wide-scale proteomic analysis in a large cohort, and it is the first cohort study with sufficient power to assess the plasma proteome's mediating and moderating effects on the link between poor CVH and dementia. Second, using diagnosis dates via record linkage rather than self-report, outcome variables were generated. Third, the UK Biobank includes a broad variety of subjects, allowing for unbiased estimates of exposure-outcome relationships via confounder correction. Potential research limitations include selection bias owing to missing data, measurement inaccuracy due to the use of self-report and ICD-10 codes, and other factors. Furthermore, as with most studies, the exact onset of dementia is unknown. Moreover, misclassification and underestimation of the incident dementia outcome is possible, even though less so for all-cause dementia compared with AD-type dementia. Although various confounders were accounted for, residual confounding is conceivable owing to the observational nature of this research. Furthermore, while prevalent dementia cases were excluded, reverse causality remains a possible explanation. We note that the use of 4-way decomposition models relies on a strong set of assumptions, notably the absence of unmeasured confounding, which we cannot explicitly rule out or experimentally evaluate (Appendix IV) (VanderWeele, 2014). Furthermore, the possibility of exposure-induced mediator-outcome confounding cannot be ruled out, and *med4way*, to the best of our knowledge, does not account for some of the innovative techniques to effect decomposition that have been proposed to reduce this possibility (Discacciati et al., 2018). Moreover, although Bonferroni correction was conducted in part of the main analyses when examining exposure-mediator strength of association, the role of chance in the four-way decomposition models was not fully accounted for, particularly in the secondary analysis including all potential plasma protein mediators or moderator followed by pathway analyses. Another possible constraint of using UK Biobank data is self-selection and healthy user bias, given considerable variations in cohort characteristics compared to the overall UK population (Swanson, 2012). However, studies that have taken this possible constraint into account have claimed that, despite this limitation, the findings are deemed to be valid and generalizable among the research participants. The similarity of our findings with those published in other comparable neuroepidemiology investigations backs up this claim (Fry et al., 2017). Similarly, the UK Biobank is not necessarily representative of the UK population, and our analyses did not possess sufficient power to be stratified by individual racial/ethnic groupings.

Finally, dementia is associated with poor CVH, a connection that is mediated by multiple plasma proteins, with GDF15 serving as the principal driver, closely followed by ADM. Future research should explore further into various characteristics of CVH in connection to dementia incidence, determining if comparable processes are implicated across these features and assessing the validity of these results in groups at highest risk for dementia in the UK.

Supplementary Material

Refer to Web version on PubMed Central for supplementary material.

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List of Appendices

Appendix I – Authors

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May A. Beydoun, PhD	<i>Laboratory of Epidemiology and Population Sciences, National Institute on Aging, NIA/NIH/IRP, Baltimore, MD, USA 21224</i>	Study concept, data acquisition, plan of analysis, data management and statistical analysis, literature search and review, write-up of parts of the manuscript, revision of the manuscript.
Hind A. Beydoun, PhD	<i>Laboratory of Epidemiology and Population Sciences, National Institute on Aging, NIA/NIH/IRP, Baltimore, MD, USA 21224</i>	Study concept, plan of analysis, assistance with data management and statistical analysis, literature search and review, write-up of parts of the manuscript, revision of the manuscript.
Yi-Han Hu	<i>Laboratory of Epidemiology and Population Sciences, National Institute on Aging, NIA/NIH/IRP, Baltimore, MD, USA 21224</i>	Study concept, plan of analysis, literature search and review, data visualization, review of the manuscript.
Christian A. Maino Vieytes, PhD	<i>Laboratory of Epidemiology and Population Sciences, National Institute on Aging, NIA/NIH/IRP, Baltimore, MD, USA 21224</i>	Plan of analysis, assistance with data management, data visualization, write-up of parts of the manuscript, literature search and review, revision of the manuscript.
Nicole Noren Hooten, PhD	<i>Laboratory of Epidemiology and Population Sciences, National Institute on Aging, NIA/NIH/IRP, Baltimore, MD, USA 21224</i>	Study concept, plan of analysis, literature search and review, write-up of parts of the manuscript, review of the manuscript.
Minkyo Song, PhD	<i>Laboratory of Epidemiology and Population Sciences, National Institute on Aging, NIA/NIH/IRP, Baltimore, MD, USA 21225</i>	Plan of analysis, literature search and review, write-up of parts of the manuscript, revision of the manuscript.
Michael F. Georgescu, MPH	<i>Laboratory of Epidemiology and Population Sciences, National Institute on Aging, NIA/NIH/IRP, Baltimore, MD, USA 21224</i>	Literature search and review, write-up of parts of the manuscript, revision of the manuscript.
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Name	Location	Contributions
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ABBREVIATIONS

AD	Alzheimer’s disease
AHA	American Heart Association
Aβ	Amyloid beta
BMI	Body mass index
CI	Confidence interval
CNS	Central nervous system
CVD	Cardiovascular disease
CVH	Cardiovascular health
GDF15	growth differentiation factor 15
ereri_cde	excess relative risk due to neither mediation nor interaction or controlled direct effect
ereri_intmed	excess relative risk due to mediated interaction or mediated interaction
ereri_intref	excess relative risk due to interaction only or interaction referent
ereri_pie	excess relative risk due to mediation only or pure indirect effect
HR	Hazard ratio
ICD-10	International Classification of Diseases, Tenth Revision
LE8	Life’s Essential 8
LS7	Life’s Simple 7
MRI	magnetic resonance imaging
OLS	Ordinary lease squares

PCA	Principal Components Analysis
pct_cde	percent of total effect that is controlled direct effect
pct_intmed	percent of total effect that is mediated interaction
pct_intref	percent of total effect that is interaction referent
pct_pie	percent of total effect that is pure indirect effect
PH	Proportional hazards
PIE	Pure Indirect Effects
PRS	Polygenic Risk Score
R	Right
Ref	Referent category
SD	Standard Deviation
SEM	Structural Equations modeling
SES	Socioeconomic status
TDI	Townsend Deprivation Index
tereri	Total excess relative risk
UK	United Kingdom
UKB-PPP	UK Biobank Pharma Proteomics Project

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HIGHLIGHTS

- Up to 28,974 UK Biobank participants were selected for this study.
- Poor cardiovascular health increased dementia risk by 11%.
- Of 11 consistent mediating plasma proteins, growth differentiation factor 15 and adrenomedullin were the strongest.
- Dementia is linked to poor CVH mediated by GDF15 and ADM among several key proteomic markers, explaining approximately 54% of the total effect.

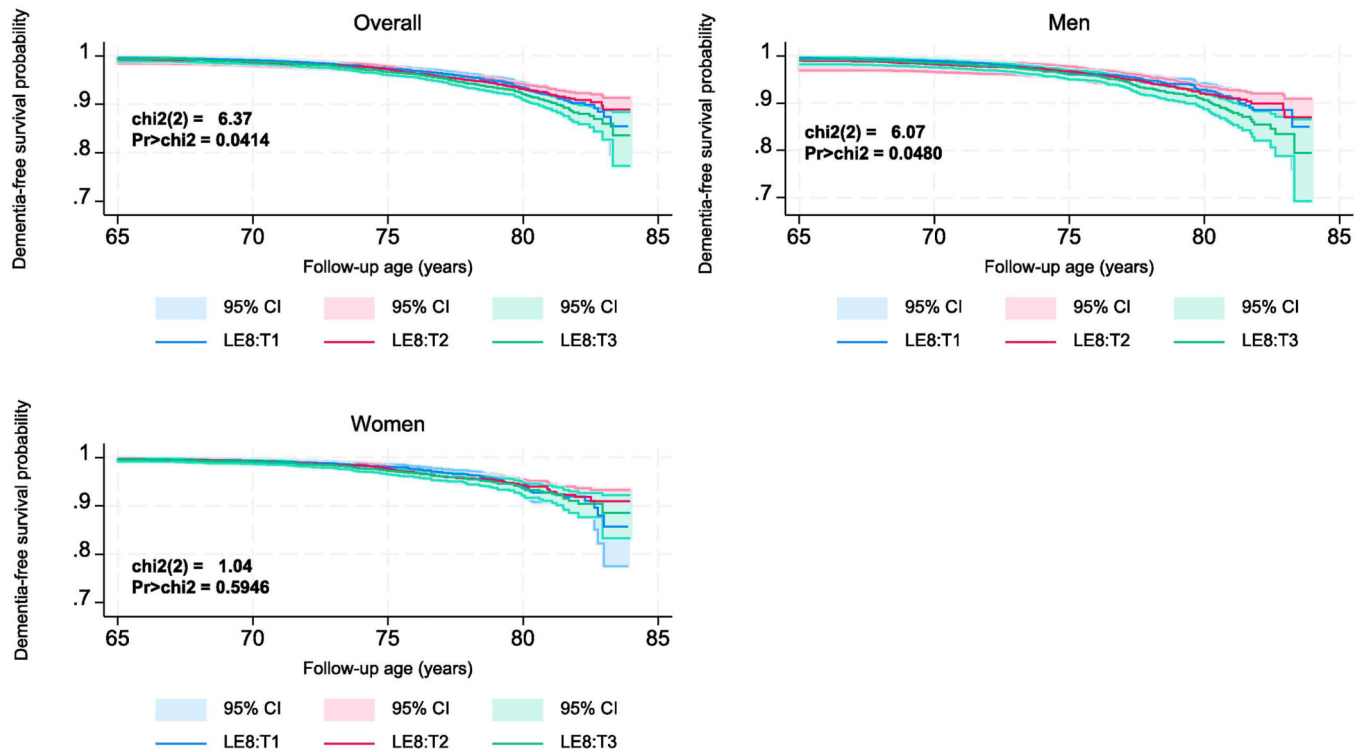


FIGURE 1. Poor cardiovascular health (CVH; reversed tertiles of LE8) vs. all-cause dementia, overall and by sex: UK biobank 2006–2021

Abbreviations: chi2=chi-square test; CI=confidence interval; KM=Kaplan Meier;

LE8=Life's Essential 8; Pr=P-values; T1=First tertile (ideal CVH); T2=Second tertile (intermediate CVH); T3=Uppermost tertile (Poor CVH); UK=United Kingdom.

Note: Chi2 refers to a log-rank test. Details are provided on github: <https://github.com/baydounm/UKB-paper11-supplementarydata>.

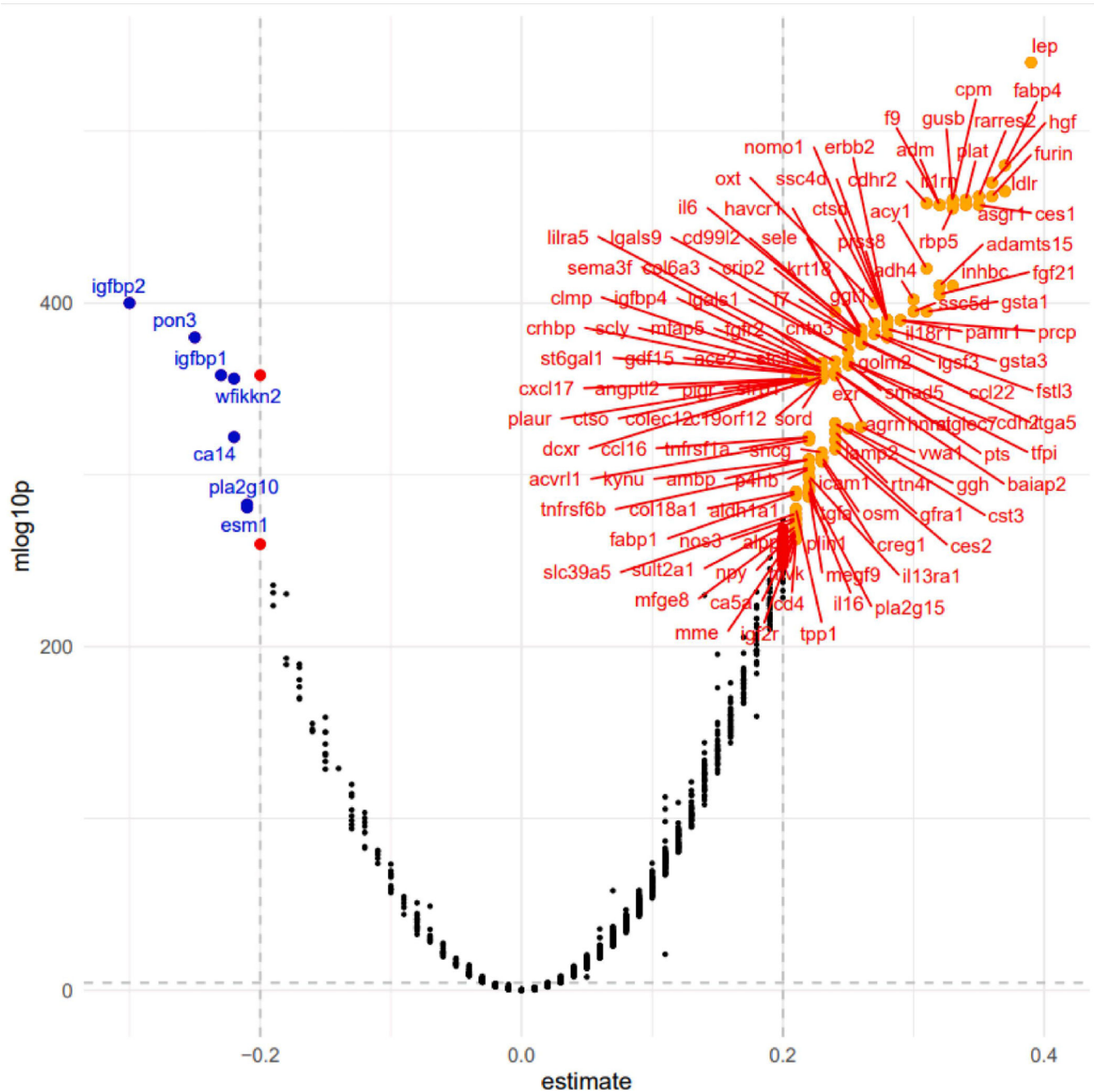
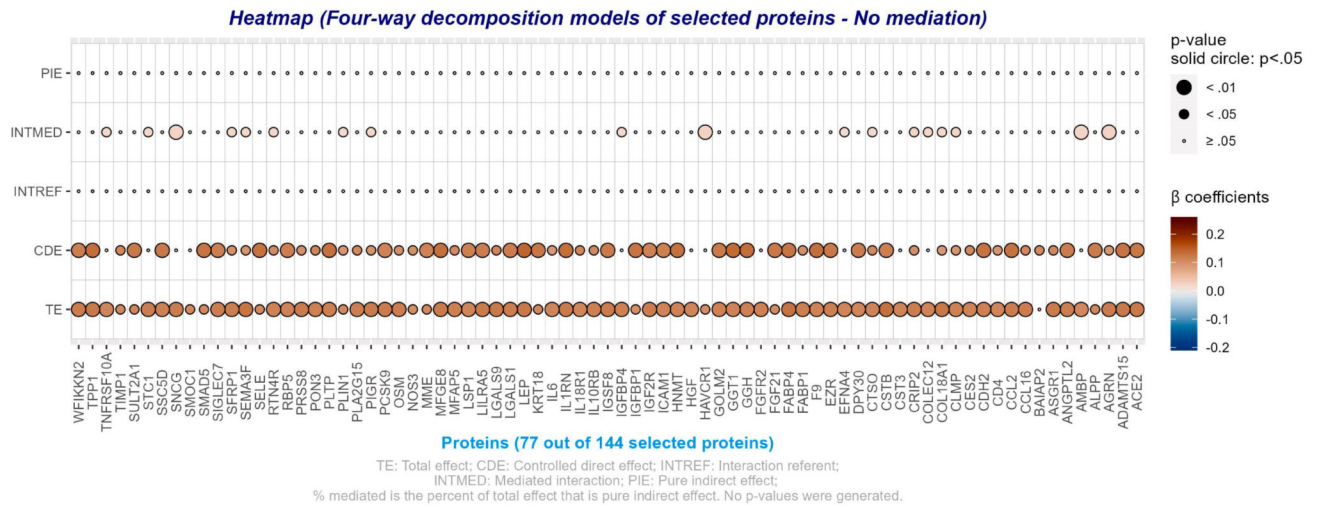
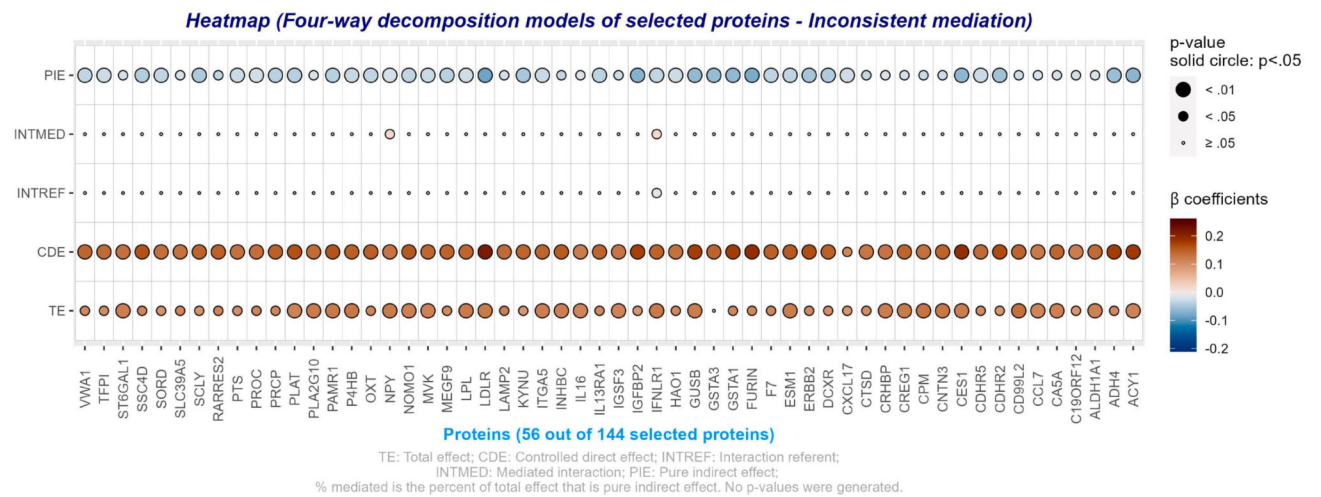


FIGURE 2. Volcano plot of plasma proteomic biomarkers in relation to LE8 reverse-coded z-scored total score: UK biobank 2006–2010

Abbreviations: See list of abbreviations for protein abbreviations; LE8=Life's Essential 8.

Note: Based on a series of multiple linear regression models, with main predictor being LE8 total score (z-scored, reverse-coded by multiplying by -1) and the outcome being each of 1,463 plasma proteomic biomarkers (Log2 transformed, z-scored). The y-axis is the predictor's associated p-value on a -Log10 scale and the X-axis is the β coefficient (effect of LE8 exposure on standardized z-scores of plasma proteomic markers) from the multiple linear regression models. An estimate with a Bonferroni corrected p-value < 0.05

and an absolute estimate >0.20 is marked by the plasma proteomic marker abbreviation (See UKB showcase URL: <https://biobank.ndph.ox.ac.uk/showcase/>). Selected proteins ($k=144$) for further mediation analysis have a corrected p -value <0.05 and a point estimate >0.20 in absolute value (red). Dashed vertical lines are set at -0.20 and $+0.20$ effect sizes. Dashed horizontal line is set at the Bonferroni corrected p -value. Details are provided on github: <https://github.com/baydounm/UKB-paper11-supplementarydata>.

(A) Heatmap for raw 4-way decomposition: PIE $p > 0.05$ (no significant pure mediation) , $k=77$ **(B) Heatmap for raw 4-way decomposition: PIE $p < 0.05$, PIE < 0 , TE $< \text{CDE}$ (inconsistent mediation), $k=56$** 

(C) Heatmap for raw 4-way decomposition: $PIE < 0.05$, $PIE > 0$, $TE > CDE$ (consistent mediation), $k=11$

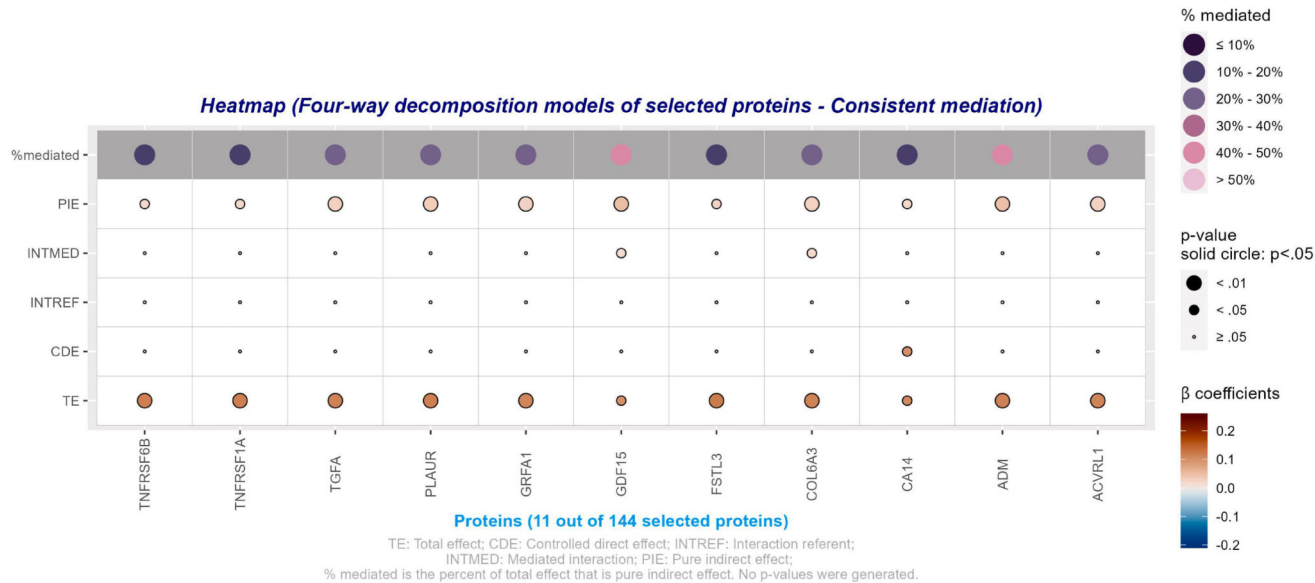


FIGURE 3. Four-way decomposition of the association between poor cardiovascular health as measured by LE8 total score (z-scored, reverse coded) and incidence of all-cause dementia by the selected plasma proteomic biomarkers (k=144): UK biobank 2006–2021

(A) Heatmap for raw 4-way decomposition: $PIE > 0.05$ (no significant pure mediation), $k=77$

(B) Heatmap for raw 4-way decomposition: $PIE < 0.05$, $PIE < 0$, $TE < CDE$ (inconsistent mediation), $k=56$

(C) Heatmap for raw 4-way decomposition: $PIE < 0.05$, $PIE > 0$, $TE > CDE$ (consistent mediation), $k=11$

Note: Mediation Effects could range from 0 to 0.5.

Abbreviations: CDE= excess relative risk due to neither mediation nor interaction or controlled direct effect; INTMED= excess relative risk due to mediated interaction or mediated interaction; INTREF= excess relative risk due to interaction only or interaction referent; PIE= excess relative risk due to mediation only or pure indirect effect; TE= Total excess relative risk; UK=United Kingdom. See supplementary Table 5 for protein abbreviations. Other Protein abbreviations are found at <https://www.ncbi.nlm.nih.gov/gene/>.

Table 1.

Study sample characteristics by sex: UK biobank 2006–2021

	Overall (N=28,974)	Men (N=13,759)	Women (N=15,215)	P _{sex}
Demographic				
Baseline age, y,	60.68 ± 0.03	61.02 ± 0.05	60.37±0.04	<0.001
Sex, % female	52.5%	--	--	
Race/ethnicity				
White	95.6%	95.6%	95.8%	(Ref)
Black	1.2%	1.1%	1.3%	0.078
South Asian	1.4%	1.6%	1.2%	0.002
Other	1.8%	1.6%	1.9%	0.014
Non-White, %	4.40%	4.31	4.48	0.47
Household size	2.223 ± 0.007	2.298 ± 0.010	2.156±0.010	<0.001
Socioeconomic				
Education				
Low	22.2%	24.2%	20.2%	<0.001
Intermediate	39.6%	34.7%	43.9%	(Ref)
High	38.2%	40.9%	35.8%	<0.001
Income				
Less than £18,000	27.0%	24.8%	29.1%	0.020
£18,000–£29,999	28.5%	27.3%	29.6%	(Ref)
£30,000–£51,999	23.7%	24.4%	23.1%	<0.001
£52,000–£100,000	16.6%	18.7%	14.6%	<0.001
greater than £100,000	4.2%	4.9%	3.6%	<0.001
Townsend Deprivation Index	−1.440 ± 0.018	−1.426± 0.026	−1.453±0.024	0.45
SES	−0.031±0.004	−0.010±0.006	−0.050±0.006	<0.001
Life's essential 8, Mean±SE				
Total score	498.8 ± 0.5	491.8 ± 0.8	505.1±0.8	<0.001
LE8z_rev	1.30e-08±0.0058748	0.0734749± 0.0084325	−0.0664437±0.0081489	<0.001
Lifestyle score	254.5 ± 0.4	248.8 ± 0.5	259.7±0.5	<0.001
Diet component	33.6 ± 0.2	28.7 ± 0.3	38.1 ± 0.3	<0.001
Physical activity component	47.7 ± 0.2	48.9 ± 0.3	46.6 ± 0.3	<0.001
Smoking component	84.4 ± 0.2	81.8 ± 0.3	86.6 ± 0.2	<0.001
Sleep component	88.9 ± 0.1	89.3 ± 0.2	88.4 ± 0.2	<0.001
Biological score	244.3 ± 0.4	243.0 ± 0.5	245.4±0.5	0.001
BMI component	68.1 ± 0.2	65.9 ± 0.2	70.1 ± 0.2	<0.001
Lipids component	49.1 ± 0.2	54.6 ± 0.3	44.0 ± 0.2	<0.001
Glucose component	89.0 ± 0.1	87.6 ± 0.2	90.2 ± 0.2	<0.001
Blood pressure component	38.1 ± 0.2	35.0 ± 0.2	41.0 ± 0.3	<0.001
AD PRS				
Tertile, %				
T1	33.3	33.2	33.5	—

	Overall (N=28,974)	Men (N=13,759)	Women (N=15,215)	P _{sex}
T2	33.3	33.2	33.4	0.94
T3	33.3	33.6	33.1	0.45
Cumulative incidence, %				
All-cause dementia	2.7%	3.2%	2.3%	<0.001
AD dementia	1.2%	1.3%	1.2%	0.34

Abbreviations. AD=Alzheimer's Disease; LE8=Life's essential 8; PRS=Polygenic Risk Score; Ref=referent category; SE=Standard Error; UK=United Kingdom

Notes: No multiple imputation was carried out in this analysis. P-value is associated with the parameter for sex in bivariate linear and multinomial logistic regression analyses, with the main outcome being a continuous or categorical characteristic, respectively. (Ref) is the referent category in the multinomial logistic regression model. Values are means±SE or percentages.

Table 2.
Poor cardiovascular health as measured by Life’s Essential 8 total score (z-scored, reverse coded) and incident dementia, overall, by sex and stratified by AD PRS tertile: UK biobank 2006–2021)

	Overall (N=28,974)		By AD PRS tertiles (N=28,801) HR with 95% CI			
	LE8 z-score, reverse-coded (LE8z_rev) HR with 95% CI	P _{LE8z_rev}	T ₁ (N= 9,601)	T ₂ (N=9,600)	T ₃ (N=9,600)	P _{LE8z_rev×ADPRStert}
Overall, N=28,974	1.11 (1.03–1.20)	0.005	1.42 (1.17–1.72)	1.14 (0.98–1.31)	1.05 (0.96–1.16)	0.006
Men, N=13,759	1.120 (1.01–1.24)	0.026	1.51 (1.17–1.93)	1.20 (0.99–1.46)	1.02 (0.89–1.16)	0.001
Women, N=15,215	1.10 (0.99–1.23)	0.088	1.25 (0.91–1.71)	1.06 (0.84–1.34)	1.10 (0.95–1.26)	0.79

Abbreviations: AD=Alzheimer’s Disease; LE8=Life’s Essential 8; PRS=Polygenic Risk Score; UK=United Kingdom

* All Cox proportional hazards models were adjusted for baseline age, sex, race/ethnicity (Non-White vs. White), household size and SES z-score. Interaction between LE8z_rev and AD PRS tertiles was tested, by including a 2-way interaction term in the main adjusted model. Values are hazard ratios with 95% CI. 1 SD of LE8z_rev is equivalent to 93.1 point reduction in LE8 total score.

Table 3.

Four-way decomposition of the association between poor cardiovascular health (LE8, z-scored, reverse coded) and all-cause dementia through selected plasma proteomic biomarkers (k=11; N_{max}=28,974, with consistent mediation): UK biobank 2006–2021

FOURWAYDECOMP	β or %	SE	z	P	LCL	UCL	PROTEIN
tereri	+0.1163	0.0423	2.75	0.006	0.0334	0.1993	tnfrsf1a
ereri_cde	+0.0845	0.0438	1.93	0.054	-0.0013	0.1703	tnfrsf1a
ereri_intref	0.0032	0.0036	0.9	0.371	-0.0038	0.0103	tnfrsf1a
ereri_intmed	0.0090	0.0079	1.14	0.254	-0.0064	0.0244	tnfrsf1a
ereri_pie	0.0197	0.0082	2.4	0.016	0.0036	0.0357	tnfrsf1a
p_cde	0.73	0.15	4.89	<0.001	0.44	1.02	tnfrsf1a
p_intref	0.03	0.03	0.85	0.395	-0.04	0.09	tnfrsf1a
p_intmed	0.08	0.07	1.08	0.281	-0.06	0.21	tnfrsf1a
p_pie	0.17	0.09	1.82	0.068	-0.01	0.35	tnfrsf1a
tereri	0.1060	0.0423	2.51	0.012	0.0231	0.1890	ca14
ereri_cde	0.0969	0.0430	2.25	0.024	0.0126	0.1813	ca14
ereri_intref	-0.0017	0.0020	-0.85	0.396	-0.0055	0.0022	ca14
ereri_intmed	-0.0064	0.0091	-0.7	0.482	-0.0243	0.0115	ca14
ereri_pie	0.0172	0.0087	1.97	0.048	0.0001	0.0343	ca14
p_cde	0.91	0.12	7.38	<0.001	0.67	1.16	ca14
p_intref	-0.02	0.02	-0.79	0.429	-0.05	0.02	ca14
p_intmed	-0.06	0.09	-0.66	0.51	-0.24	0.12	ca14
p_pie	0.16	0.10	1.57	0.117	-0.04	0.36	ca14
tereri	0.0978	0.0418	2.34	0.019	0.0160	0.1797	gdf15
ereri_cde	0.0259	0.0430	0.6	0.547	-0.0584	0.1102	gdf15
ereri_intref	0.0107	0.0060	1.8	0.072	-0.0010	0.0224	gdf15
ereri_intmed	0.0142	0.0069	2.05	0.04	0.0006	0.0277	gdf15
ereri_pie	0.0470	0.0081	5.78	<0.001	0.0311	0.0630	gdf15
p_cde	0.26	0.33	0.79	0.429	-0.39	0.92	gdf15
p_intref	0.11	0.08	1.4	0.162	-0.04	0.26	gdf15
p_intmed	0.14	0.09	1.64	0.100	-0.03	0.32	gdf15
p_pie	0.48	.2181267	2.20	0.027	0.05	0.91	gdf15
tereri	0.1194	0.0424	2.81	0.005	0.0363	0.2025	fstl3
ereri_cde	0.0792	0.0443	1.79	0.074	-0.0076	0.1659	fstl3
ereri_intref	0.0037	0.0037	1.02	0.307	-0.0034	0.0109	fstl3
ereri_intmed	0.0135	0.0103	1.31	0.191	-0.0067	0.0336	fstl3
ereri_pie	0.0230	0.0109	2.11	0.035	0.0017	0.0444	fstl3
p_cde	0.66	0.18	3.78	<0.001	0.32	1.01	fstl3
p_intref	0.03	0.03	0.98	0.329	-0.03	0.09	fstl3
p_intmed	0.11	0.09	1.24	0.214	-0.06	0.29	fstl3
p_pie	0.19	0.11	1.69	0.092	-0.03	0.42	fstl3
tereri	0.1128	0.0415	2.72	0.007	0.0315	0.1942	col6a3

FOURWAYDECOMP	β or %	SE	z	P	LCL	UCL	PROTEIN
ereri_cde	0.0629	0.0426	1.48	0.139	-0.0205	0.1463	col6a3
ereri_intref	0.0076	0.0046	1.64	0.102	-0.0015	0.0166	col6a3
ereri_intmed	0.0183	0.0086	2.13	0.033	0.0015	0.0352	col6a3
ereri_pie	0.0240	0.0092	2.6	0.009	0.0059	0.0421	col6a3
p_cde	0.56	0.20	2.8	0.005	0.17	0.95	col6a3
p_intref	0.07	0.05	1.45	0.147	-0.02	0.16	col6a3
p_intmed	0.16	0.09	1.83	0.067	-0.01	0.34	col6a3
p_pie	0.21	0.11	1.88	0.061	-0.01	0.43	col6a3
tereri	0.1163	0.0425	2.74	0.006	0.0330	0.1996	plaur
ereri_cde	0.0733	0.0439	1.67	0.095	-0.0127	0.1593	plaur
ereri_intref	0.0037	0.0050	0.74	0.46	-0.0061	0.0135	plaur
ereri_intmed	0.0076	0.0082	0.94	0.349	-0.0084	0.0237	plaur
ereri_pie	0.0316	0.0089	3.56	<0.001	0.0142	0.0491	plaur
p_cde	0.63	0.18	3.56	<0.001	0.28	0.98	plaur
p_intref	0.03	0.04	0.72	0.474	-0.06	0.12	plaur
p_intmed	0.07	0.07	0.91	0.36	-0.08	0.21	plaur
p_pie	0.27	0.12	2.19	0.029	0.03	0.52	plaur
tereri	0.1132	0.0418	2.7	0.007	0.0312	0.1951	adm
ereri_cde	0.0433	0.0436	0.99	0.321	-0.0422	0.1288	adm
ereri_intref	0.0064	0.0057	1.12	0.265	-0.0048	0.0176	adm
ereri_intmed	0.0173	0.0126	1.38	0.169	-0.0073	0.0419	adm
ereri_pie	0.0463	0.0136	3.4	0.001	0.0196	0.0729	adm
p_cde	0.38	0.27	1.4	0.162	-0.15	0.92	adm
p_intref	0.06	0.05	1.09	0.276	-0.04	0.16	adm
p_intmed	0.15	0.11	1.36	0.173	-0.07	0.37	adm
p_pie	0.41	0.19	2.15	0.031	0.04	0.78	adm
tereri	0.1085	0.0417	2.6	0.009	0.0267	0.1902	gfra1
ereri_cde	0.0672	0.0427	1.57	0.116	-0.0165	0.1509	gfra1
ereri_intref	0.0047	0.0043	1.08	0.28	-0.0038	0.0132	gfra1
ereri_intmed	0.0119	0.0088	1.35	0.176	-0.0053	0.0291	gfra1
ereri_pie	0.0247	0.0091	2.7	0.007	0.0068	0.0426	gfra1
p_cde	0.62	0.19	3.25	0.001	0.25	0.99	gfra1
p_intref	0.04	0.04	1.03	0.302	-0.04	0.13	gfra1
p_intmed	0.11	0.08	1.3	0.195	-0.06	0.27	gfra1
p_pie	0.23	0.12	1.86	0.063	-0.01	0.47	gfra1
tereri	0.1105	0.0425	2.6	0.009	0.0272	0.1937	acvrl1
ereri_cde	0.0755	0.0441	1.71	0.087	-0.0109	0.1619	acvrl1
ereri_intref	0.0029	0.0040	0.72	0.472	-0.0049	0.0107	acvrl1
ereri_intmed	0.0072	0.0080	0.9	0.366	-0.0084	0.0228	acvrl1
ereri_pie	0.0249	0.0085	2.92	0.003	0.0082	0.0416	acvrl1
p_cde	0.68	0.17	4.03	<0.001	0.35	1.02	acvrl1
p_intref	0.03	0.04	0.69	0.49	-0.05	0.10	acvrl1

FOURWAYDECOMP	β or %	SE	z	P	LCL	UCL	PROTEIN
p_intmed	0.07	0.07	0.87	0.385	-0.08	0.21	acvr11
p_pie	0.23	0.12	1.96	0.05	0.00	0.45	acvr11
tereri	0.1159	0.0423	2.74	0.006	0.0330	0.1989	tnfrsf6b
ereri_cde	0.0776	0.0430	1.81	0.071	-0.0066	0.1618	tnfrsf6b
ereri_intref	0.0062	0.0045	1.38	0.166	-0.0026	0.0149	tnfrsf6b
ereri_intmed	0.0156	0.0081	1.92	0.054	-0.0003	0.0315	tnfrsf6b
ereri_pie	0.0165	0.0082	2.01	0.045	0.0004	0.0327	tnfrsf6b
p_cde	0.67	0.16	4.09	<0.001	0.35	0.99	tnfrsf6b
p_intref	0.05	0.04	1.29	0.196	-0.03	0.13	tnfrsf6b
p_intmed	0.13	0.08	1.74	0.082	-0.02	0.29	tnfrsf6b
p_pie	0.14	0.09	1.61	0.107	-0.03	0.32	tnfrsf6b
tereri	0.1126	0.0425	2.65	0.008	0.0293	0.1959	tgfa
ereri_cde	0.0811	0.0428	1.89	0.058	-0.0029	0.1651	tgfa
ereri_intref	0.0011	0.0043	0.26	0.796	-0.0073	0.0095	tgfa
ereri_intmed	0.0033	0.0083	0.4	0.691	-0.0129	0.0195	tgfa
ereri_pie	0.0271	0.0083	3.28	0.001	0.0109	0.0433	tgfa
p_cde	0.72	0.15	4.76	<0.001	0.42	1.02	tgfa
p_intref	0.01	0.04	0.26	0.794	-0.06	0.08	tgfa
p_intmed	0.03	0.07	0.4	0.687	-0.11	0.17	tgfa
p_pie	0.24	0.12	2.07	0.039	0.01	0.47	tgfa

Abbreviations: ereri_cde= excess relative risk due to neither mediation nor interaction or controlled direct effect; ereri_intmed= excess relative risk due to mediated interaction or mediated interaction; ereri_intref= excess relative risk due to interaction only or interaction referent; ereri_pie= excess relative risk due to mediation only or pure indirect effect; LE8=Life's Essential 8; p_cde=proportion of total effect that is controlled direct effect; p_intmed=proportion of total effect that is mediated interaction; p_intref=proportion of total effect that is interaction referent; p_pie=proportion of total effect that is pure indirect effect; tereri= Total excess relative risk; UK=United Kingdom. Names of the genes/proteins can be found at: <https://www.ncbi.nlm.nih.gov/gene>. Tereri and ereri_cde are interpreted as Loge(hazard ratios). 1 SD of LE8z_rev is equivalent to 93.1 point reduction in LE8 total score.